Coding Basics ⌨

Brownie Coding for Good Badge — Girls Build Bots

Presented by your robotics mentors!

We’re not just coding today 💚

We’re coding **for good**.

Code isn’t just for games — it’s a tool you can use to make something **better for someone else**.

Question for you: who could you help with a computer?

What is an ALGORITHM? 📋

An **algorithm** is just a **list of steps in the right order**.

You already use them every day: - Brushing your teeth 🪥 - Planting a seed 🌱 - Making a snack 🥣

Game Time: The Algorithm Snack 🥣

Your mentor is now a ROBOT. You give the steps!

- "Make it good" → *the robot FREEZES* ❄
- Milk before the cup → *a happy mess!* 💦

Computers follow your steps **exactly**.

So the order matters.

Unplugged: Algorithm Cards 

With a partner:

1. Put your picture cards in the **right order**
2. Trade cards with your partner
3. **Act out** their algorithm exactly
4. A step missing or out of order? That's a **BUG**! 

Bugs are GOOD! 

When a program doesn’t work, that’s called a **bug**.
Finding a bug means you’re doing **real programming**.
Fixing it is called **debugging**.

Now let’s code on a screen! 💻

Follow your mentor, one block at a time:

1. Make a character — that’s called a **sprite** ✨
2. Make it **move** across the screen
3. Make it **say hello** or play a sound 🔊

Finished early? Change your character and add a second action!

Where’s the code hiding? 🔎

Code is all around you. Where do you think it’s hiding?

- Traffic lights 🚦
- Video games 🎮
- The microwave 🍿
- A talking toy 🧸

Algorithms are everywhere.

Robot moment 

Our competition robot runs on code too.

The same kind of **step-by-step instructions** you just wrote — they just control **motors** instead of a sprite.

Show and Tell ✨

Show us **ONE thing** your program does!

You did it! 🎉

Today you: - ✅ Learned that an algorithm is a list of steps in order - ✅ Found and fixed a bug - ✅ Made a sprite move and react - ✅ Found code hiding in everyday life

You are a programmer now!

Keep going at home! 🏠

- **MakeCode Arcade:** arcade.makecode.com (free!)
- **ScratchJr** app — make your own stories and games
- Ask your troop leader about the next Coding for Good badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.